ASSOCIATED POINTS OF A COHERENT MODULE ARE FINITE OVER A QUASI-COMPACT OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof is a translation: the geometric statement is reduced to the algebra fact by four named steps written out in the proof, and the algebra fact itself is delegated to the Algebra chapter; that delegation is recorded here for the reader.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- A finite module over a Noetherian ring has finitely many associated primes. Algebra, Lemma algebra-lemma-finite-ass, used at proof lines 158-160. Is the algebra fact the reduction lands on.
- An affine scheme of a locally Noetherian scheme is the spectrum of a Noetherian ring. Properties, Lemma properties-lemma-locally-Noetherian, used at proof lines 161-163. Supplies the Noetherian hypothesis of the algebra fact.
- A coherent module on a Noetherian affine is a finite module. Cohomology of Schemes, Lemma coherent-lemma-coherent-Noetherian, used at proof lines 163-165. Turns F into a finite module.
- Associated points on an affine scheme are associated primes. Lemma lemma-associated-affine-open, used at proof lines 165-166. Completes the translation.

Source reference. Original lemma 00LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 782–782. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01XZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2538–2538. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02OK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3519–3519. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01OW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2209–2209. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian scheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Then $Ass(\mathcal{F}) \cap U$ is finite for every quasi-compact open $U \subset X$.*

Proof. This is true because the set of associated primes of a finite module over a Noetherian ring is finite, see Algebra, Lemma 00LC. To translate from schemes to algebra use that U is a finite union of affine opens, each of these opens is the spectrum of a Noetherian ring (Properties, Lemma 01OW), $\mathcal{F}$ corresponds to a finite module over this ring (Cohomology of Schemes, Lemma 01XZ), and finally use Lemma 02OK. □